

VolFormer: Explore More Comprehensive Cube Interaction for Hyperspectral Image Restoration and Beyond

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Abstract

Capitalizing on the talent of self-attention in capturing non-local features, Transformer architectures have exhibited remarkable performance in single hyperspectral image restoration. For hyperspectral images, each pixel is located in the hyperspectral image cubes with a large spectral dimension and two spatial dimensions. Although uni-dimensional self-attention, like channel self-attention or spatial self-attention, builds long-range dependencies in spectral or spatial dimensions, they lack more comprehensive interactions across dimensions. To tackle the above drawback, we propose a VolFormer, a volumetric self-attention embedded Transformer network for single hyperspectral image restoration. Specifically, we propose volumetric self-attention (VolSA), which extends the interaction from 2D flat to 3D cube. VolSA can simultaneously model token interaction in the 3D cube, mining the potential correlations between the hyperspectral image cube. An attention decomposition form is proposed to reduce the computational burden of modeling volumetric information. In practical terms, VolSA adapts double similarity matrixes in spatial and channel dimensions to implicitly model 3D context information while transforming the complexity from cubic to quadratic. Additionally, we introduce the explicit spectral location prior to enhance the proposed self-attention. This property allows the target token to perceive global spectral information while simultaneously assigning different levels of attention to tokens at varying wavelength bands. Extensive experiments demonstrate that VolFormer achieves record-high performance on hyperspectral image super-resolution, denoise and classification benchmarks. The source code is available at <https://github.com/yudadabing/VolFormer>.

1. Introduction

Hyperspectral image (HSI) is referred to as an image cube, which has a large spectral dimension besides spatial dimension. Large spectral dimension provide ample and detailed spectral information to identify and distinguish objects. Therefore, hyperspectral images have been widely used in various tasks. [33, 37]. Due to the physical limitations of imaging sensors and long imaging time, the captured hyperspectral images maintain low resolution and introduce noise. Thus, hyperspectral image restoration is mainly categorized into hyperspectral image super-resolution and hyperspectral image denoise, which aims to restore a high-quality image from the degraded observation.

Deep learning methods based on data-driven have explicit performance over conventional prior-based methods in most computer vision tasks. Thus, convolutional neural networks (CNNs) have been used in HSI restoration tasks and demonstrate the powerful spatial-spectral feature extraction ability. However, the local feature learning characteristic of prevailing CNN methods shows limitations in modeling non-local correlation. Extracting non-local features is equally significant in high-dimensional data of hyperspectral images. Vision Transformer utilizes self-attention to capture long-range dependencies and provide powerful local and non-local feature representations. Recent work [56, 62] makes an effort to tailor the vision Transformer for hyperspectral image restoration tasks. These Transformers exploit the self-attention mechanisms (*i.e.*, channel self-attention and spatial self-attention) to extract the non-local features in spatial and spectral dimensions and significantly improve visual quality and quantitative metrics. However, in channel self-attention (CSA) or spatial self-attention (SSA), the three-dimensional (3D) context information is mapped to a two-dimensional (2D) similarity matrix, where space or channel compression. Different from traditional visible images with R-G-B channels, hy-

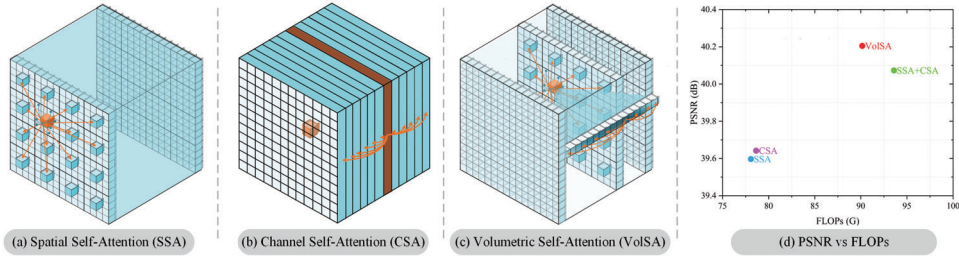


Figure 1. The interaction style of typical self-attention mechanisms and the PSNR vs FLOPs. The conventional self-attention mechanisms, including (a) spatial self-attention and (b) channel self-attention, extract the non-local features in two-dimensions flat. Our proposed (c) volumetric self-attention models the long-range dependency in a 3D cube and achieve trade-off between computational cost and performance (d) for super-resolution task.

perspectival images are high-dimensional data with numerous channels (hundreds or thousands), and each pixel is related to the other pixels located in the cube. The space or channel information compression and the lack of joint analysis between the spectrum and spatial information greatly limit the performance of the Transformer in hyperspectral image restoration.

To exploit the full potential of Transformer and pursue higher performance, in this paper, we propose the volumetric self-attention (VolSA) mechanism, which extends the interaction into three-dimensional space and offers higher-order receptive field information. As shown in Fig. 1, unlike uni-dimensional interactions (*e.g.*, channel self-attention and spatial self-attention), the volumetric self-attention aims to model more comprehensive interaction; the target token has made the response for each position of the cube. Meanwhile, in our work, the spectrum location prior is introduced to improve the efficiency of feature interaction. Specifically, since hyperspectral imaging systems acquire images at a nominal spectral resolution, HSIs have a continuous, narrowband spectrum. The continuous spectral channels represent that the spectral correlations are closely related to wavelength. That is, the closer the spectral channels are to each other, the more similar their spectral wavelengths and the reflectance information they capture will be. However, the traditional self-attention’s inherent structure lacks explicit location priors. In our volumetric self-attention, the explicit spectrum location prior is added to 2D similarity matrixes in the spectral domain, *i. e.*, for a target token, the closer the spectral channels are, the greater the degree of correlation. Self-attention’s quadratic complexity leads to significant computational costs when extending the pixel interaction to the cube. To relieve the computational costs, in the proposed volumetric self-attention, we follow the volume of cube calculation paradigm. The cube attention is respectively projected to generate 2D similarity matrixes in spatial and spectral dimensions. Then, we compute the cross-covariance across two 2D similarity matrixes to obtain the attention map, which records the poten-

tial attention relationship between the target token and all locations in the cube. Thus the attention complexity from cubic $\mathcal{O}(H^2W^2C^2)$ (for hyperspectral images of $H \times W$ pixels with C channels.) to quadratic $\mathcal{O}(H^2W^2 + C^2)$. As shown in Fig. 1(d), our VolSA obtains significant PSNR gains compared to SSA and CSA with a low complexity increase. Positively, our VolSA maintains lower FLOPs and fewer parameters than combined SSA and CSA.

Our contributions can be summarized as follows:

- We propose the VolFormer for hyperspectral image restoration. Extensive experimental results on hyperspectral super-resolution and denoise show that our VolFormer obtains notable performance in both objective metrics and subjective visual quality.
- We have tailored the volumetric self-attention for the hyperspectral images, which extends the long-range dependency into three-dimensional space and maintains the quadratic complexity.
- Considering the characteristics of the continuous spectrum, we introduce an explicit spectrum location prior to the volumetric self-attention, which employs a bidirectional decay matrix to describe the correlation of spectral channels.

2. Related Work

Hyperspectral image restoration mainly includes hyperspectral denoising and super-resolution, with a wide range of applications. In the early years, prior knowledge-based approaches [10, 14–17, 20, 22, 52] were dominant due to their ability to model the intrinsic properties of hyperspectral images. Recently, benefitting from the superior performance of CNNs in many computer vision tasks, CNNs have been introduced into restoration tasks for single hyperspectral images and have numerous variations [7, 18, 19, 23, 29, 38, 50, 60]. Since hyperspectral image contains abundant spectral information, 3D CNNs [11, 45] and mixed 2D/3D convolutional networks [26, 27] are designed to reconstruct the HSI, which extracts the spectral

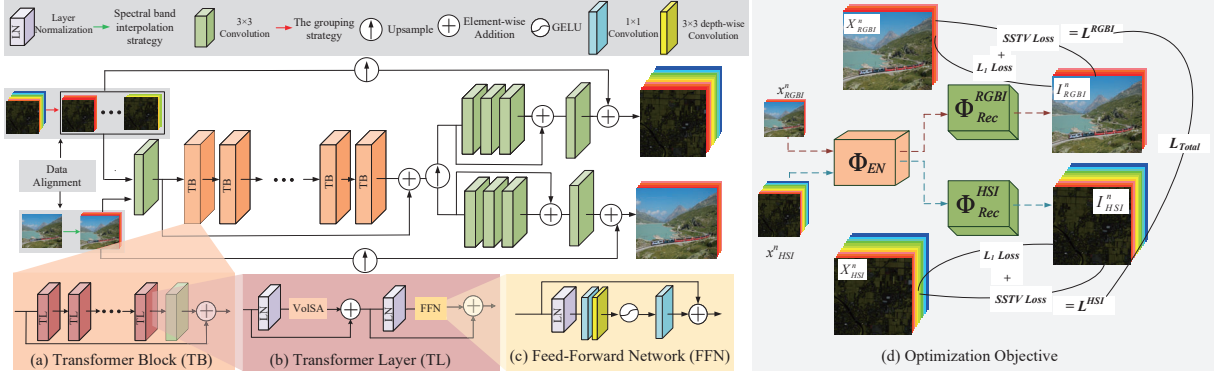


Figure 2. The architecture of VolFormer for hyperspectral image restoration. Our VolFormer mainly consists of residual in residual design for (a) Transformer blocks and (b) Transformer layers; (c) the feed-forward network (FFN) is used incorporating with volumetric self-attention (VolSA) and tends to activate local signals; (d) a visual exhibition of the loss functions.

information with expensive computational cost.

Transformer has been an effective alternative to CNNs in vision applications, *e.g.*, image recognition [8], segmentation [46], and object detection [5]. The vision Transformer model (ViT) proposed by Dosovitskiy et al. [8] applies a pure Transformer directly to sequences of image patches to classify the full image and achieves state-of-the-art performance. Naturally, Transformer has been developed to address low-level vision problems, such as image restoration. Liang et al. [30] proposed the SwinIR model for image restoration based on the Swin Transformer to apply self-attention within local image regions. Restormer [59] captures long-range channel interactions and achieves pleasing results on several image restoration tasks. Transformer’s superior ability to capture long-term information and relationships inspires its application in the field of hyperspectral image restoration [24, 25, 56, 62] and reconstruction [4, 44]. The previous works aim to calculate self-attention along the spectral and spatial dimensions to capture long-range spectrum dependencies and explore intrinsic similarity characteristics. However, existing Transformer-based methods lack the joint analysis between spectral and spatial dimensions.

3. Proposed Method

3.1. Architecture

As shown in Fig. 2, following the previous work [56], our VolFormer is also trained on heterogeneous datasets simultaneously. Specifically, in this paper, we select the RGBI dataset as the auxiliary dataset and train our VolFormer on both the RGBI and the HSI datasets. The overall structure of our VolFormer consists of three parts: shallow feature extraction, deep feature extraction, and image reconstruction.

Shallow Feature Extraction. Initially, given an HSI dataset $\Omega_{HSI} = \{x_{HSI}^i, X_{HSI}^i\}_{i=1}^{N_{HSI}}$ and RGBI dataset

$\Omega_{RGBI} = \{x_{RGBI}^i, X_{RGBI}^i\}_{i=1}^{N_{RGBI}}$, where $N_{RGBI} \gg N_{HSI}$. $x_{HSI} \in \mathbb{R}^{h \times w \times D}$ and $x_{RGBI} \in \mathbb{R}^{h \times w \times 3}$ represent the degraded samples, while $X_{HSI} \in \mathbb{R}^{H \times W \times D}$ and $X_{RGBI} \in \mathbb{R}^{H \times W \times 3}$ are the high-quality counterparts. W, H, w and h respectively denote the width and height of the desired image and degraded image, D is the number of HSI bands. We first utilize data alignment operation (see Appendix 6) to process the samples in heterogeneous datasets. The generated RGBI dataset $\bar{\Omega}_{RGBI} = \{\bar{x}_{RGBI}^i, \bar{X}_{RGBI}^i\}_{i=1}^{N_{RGBI}}$ and HSI dataset $\bar{\Omega}_{HSI} = \{\bar{x}_{HSI}^i, \bar{X}_{HSI}^i\}_{i=1}^{N_{HSI}}$ keep the same spectral bands, where $\bar{x}_{RGBI} \in \mathbb{R}^{h \times w \times S}$, $\bar{X}_{RGBI} \in \mathbb{R}^{H \times W \times S}$ and $\bar{x}_{HSI} \in \mathbb{R}^{h \times w \times S}$, $\bar{X}_{HSI} \in \mathbb{R}^{H \times W \times S}$. Our VolFormer applies a 3×3 convolutional layer to extract shallow feature maps $F_{SF}^{HSI} \in \mathbb{R}^{h \times w \times L}$ and $F_{SF}^{RGBI} \in \mathbb{R}^{h \times w \times L}$, where L is the number of channels of the shallow feature.

Deep Feature Extraction. The shallow features are transported to the shared encoder Φ_{EN} to extract deep features $F_{DF}^{HSI} \in \mathbb{R}^{h \times w \times L}$ and $F_{DF}^{RGBI} \in \mathbb{R}^{h \times w \times L}$. The shared encoder is stacked by multiple Transformer blocks (TBs), with the total number b_1 . Meanwhile, the residual strategy is employed in the shared encoder to ensure training stability. Each Transformer block contains l_1 of Transformer layers (TLs). As depicted in Fig. 2(b), each Transformer layer contains the proposed volumetric self-attention and feed-forward network, respectively. Volumetric self-attention is the core component of our VolFormer. A detailed explanation of volumetric self-attention in Sec. 3.2. The feed-forward network comprises two convolutional layers with a kernel size of 1×1 , a depth-wise convolutional layer with a kernel size of 3×3 , and the GELU activation layer. A convolution layer is introduced at the end of the Transformer blocks to refine features extracted from the Transformer layers.

Image Reconstruction. Image reconstruction Φ_{Rec} aims

to map the aggregated deep features, shallow features and degraded image to the desired high-quality output. The deep feature and shallow feature are first aggregated and up-sampled. Then, we use three 3×3 convolutional layers with the residual connection to generate the decoded feature. The convolutional layers preserve the domain gap between the heterogeneous datasets and decode the deep spatial-spectral features. Finally, the enhanced feature and upsampled degraded image are fused, generating the reconstructed high-quality result.

Optimization Objective. The L_1 loss and the spatial-spectral total variation (SSTV) loss [19] are applied to optimize the parameter distribution of VolFormer. As shown in Fig. 2(d), the overall loss of our restoration task contains the losses of the RGBI restoration task and the HSI restoration task. **More details are presented in Appendix 7.**

3.2. Volumetric Self-Attention

The RGB image is similar to a sandwich with triple-thin layers corresponding to discrete R-G-B spectral channels. While the hyperspectral image can be seen as a pumpnickel with continuous spectra. Existing self-attention paradigms, including spatial self-attention and channel self-attention, only model non-local correlation for two-dimensional slices that are difficult to mine comprehensive correlations hidden in the hyperspectral image cube. Thus, we propose volumetric self-attention (VolSA), a novel three-dimensional self-attention paradigm. VolSA employs the two 2D similarity matrixes corresponding to spectral-wise projection and spatial-wise projection to implicitly build the long-range dependencies in a cube with the relay communication way.

Spectral-wise Projection. As shown in Fig. 3, we first compute cross-covariance across channels to generate a spectral-wise projection matrix encoding the global signal in the spectral domain. Given a hyperspectral image $X_{in} \in \mathbb{R}^{H \times W \times C}$, where the W and H are the width and height of the input, and C denotes the channel number. By applying 1×1 point-wise convolutions W_P followed by 3×3 depth-wise convolutions W_D , we first generate *query* $Q_{spe} \in \mathbb{R}^{C \times HW}$, *key* $K_{spe} \in \mathbb{R}^{C \times HW}$ and *value* $V_{spe} \in \mathbb{R}^{C \times HW}$ from the input X_{in} . The project process can increase the representation ability, it is defined as

$$Q_{spe} = W_P^Q W_D^Q X_{in}, \quad K_{spe} = W_P^K W_D^K X_{in}, \quad V_{spe} = W_P^V W_D^V X_{in}. \quad (1)$$

Then, we apply dot-product interaction on Q_{spe} and K_{spe} to generate the spectral-wise projection matrix $A_{spe} \in \mathbb{R}^{C \times C}$,

$$A_{spe} = \text{Softmax} \left(\frac{Q_{spe} \cdot K_{spe}}{\varepsilon} + B \right), \quad (2)$$

where ε is a learnable parameter to reweight the dot product of Q_{spe} and K_{spe} before applying the softmax function.

B is the learnable relative positional encoding.

Traditional Transformers lack location prior. The retention mechanism [41] for sequence modeling has brought temporal prior to the language model. Inspired by the retention mechanism [9, 41], we attempt to introduce the spectrum location prior, which describes the spectral correlation of the spectrum based on the spectrum wavelength. For HSI with a continuous, narrowband spectrum, the location distance of the spectrum is related to the interspectral similarity. Each spectrum has a bidirectional decay that attends to the preceding spectrum and those following it. Hence, we broaden the temporal prior to a bidirectional form to model the bidirectional spectrum correlation decay,

$$D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ \gamma^{m-n}, & n < m \end{cases} \quad (3)$$

where D_{nm} is a matrix that combines causal masking and exponential spectrum correlation decay along relative distance.

Then, the original spectral-wise projection matrix is incorporated with the spectrum location prior. The enhanced spectral-wise projection matrix A_{spe} can be expressed as

$$A_{spe} = \text{Softmax} \left(\frac{Q_{spe} \cdot K_{spe}}{\varepsilon} \odot D + B \right). \quad (4)$$

To enhance the representation ability of the projection matrix, we attempt to project information from different representation subspaces at different positions. We split Q_{spe} , K_{spe} , V_{spe} into g heads: $Q_{spe} = [Q_{spe}^1, \dots, Q_{spe}^g]$, $K_{spe} = [K_{spe}^1, \dots, K_{spe}^g]$ and $V_{spe} = [V_{spe}^1, \dots, V_{spe}^g]$, where all components are in $\mathbb{R}^{\frac{C}{g} \times HW}$. Therefore, the spectral-wise projection matrix A_{spe} for the i -th heads is defined as

$$A_{spe}^i = \text{Softmax} \left(\frac{Q_{spe}^i \cdot K_{spe}^i}{\varepsilon} \odot D + B \right). \quad (5)$$

The latent output Y_{spe}^i for the i -th heads is defined as

$$Y_{spe}^i = A_{spe}^i \cdot V_{spe}^i. \quad (6)$$

Finally, we obtain the latent feature $Y_{spe} \in \mathbb{R}^{C \times HW}$ by reshaping and concatenating all Y_{spe}^i . The process is formulated as

$$Y_{spe} = H_{cat} (Y_{spe}^1, \dots, Y_{spe}^g), \quad (7)$$

where H_{cat} denotes the channel concatenation operation.

Spatial-wise Projection. Spatial-wise projection aims to calculate the token interactions in spatial dimension and generates a spatial-wise matrix modeling the long-range dependencies. As illustrated in Fig. 3, X_{in} is first linearly

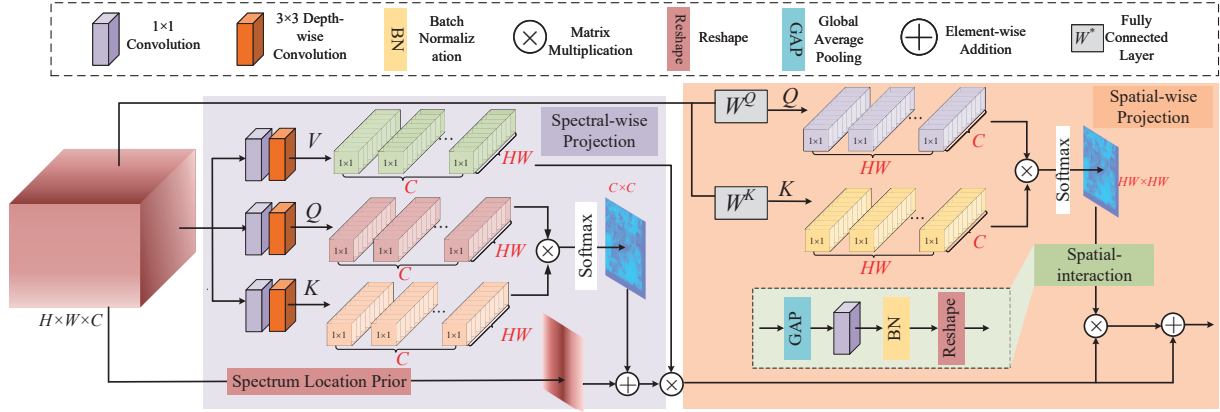


Figure 3. Overall architecture of Volumetric Self-Attention. The VolSA exploits the spectral-wise projection and spatial-wise projection to implicitly model the cube attention.

Scale	Method	CAVE						Pavia					
		SAM	CC	ERGAS	RMSE	MPSNR	MSSIM	SAM	CC	ERGAS	RMSE	MPSNR	MSSIM
×4	Bicubic	4.176	0.9868	5.272	0.0212	34.721	0.9303	7.243	0.9079	7.1571	0.0328	31.491	0.8092
	GDRRN [29]	3.726	0.9927	3.735	0.0155	37.687	0.9525	6.728	0.9258	6.5415	0.0284	32.621	0.8586
	MCNet [26]	3.412	0.9843	4.222	0.0146	37.870	0.9540	7.406	0.9236	6.5302	0.0295	32.380	0.8477
	ERCSR [27]	3.273	0.9847	4.153	0.0144	38.009	0.9553	7.039	0.9256	6.3574	0.0291	32.543	0.8507
	SSPSR [19]	3.360	0.9930	3.543	0.0146	38.302	0.9566	6.828	0.9316	6.3500	0.0277	32.783	0.8608
	HSISR [23]	3.319	0.9945	3.204	0.0131	39.060	0.9618	7.208	0.9325	6.3078	0.0288	32.754	0.8673
	DSTrans [56]	3.169	0.9953	2.861	0.0118	40.073	0.9659	7.174	0.9327	6.3847	0.0279	32.792	0.8626
	ESSA [62]	3.261	0.9940	3.299	0.0136	39.025	0.9594	7.178	0.9371	6.2807	0.0267	32.806	0.8714
	Ours	3.083	0.9953	2.856	0.0117	40.205	0.9663	6.398	0.9409	6.0690	0.0263	32.925	0.8721
×8	Bicubic	5.896	0.9666	8.435	0.0346	30.206	0.8494	8.825	0.8063	10.528	0.0476	28.190	0.6290
	GDRRN [29]	5.858	0.9731	7.346	0.0307	31.430	0.8709	8.891	0.8153	10.217	0.0463	28.349	0.6423
	MCNet [26]	5.407	0.9695	6.573	0.0278	32.417	0.8873	9.815	0.8175	10.338	0.0460	28.382	0.6478
	ERCSR [27]	5.210	0.9630	6.440	0.0267	32.602	0.8901	10.232	0.8096	10.442	0.0473	28.226	0.6317
	SSPSR [19]	4.722	0.9800	6.050	0.0257	33.217	0.8936	10.396	0.8194	10.577	0.0458	28.372	0.6322
	HSISR [23]	5.108	0.9821	5.971	0.0246	34.096	0.9101	9.294	0.8170	10.679	0.0457	28.326	0.6591
	DSTrans [56]	4.623	0.9831	5.723	0.0239	34.797	0.9167	8.958	0.8172	10.289	0.0459	28.396	0.6659
	ESSA [62]	6.766	0.9787	6.788	0.0274	32.807	0.8946	8.892	0.8171	10.349	0.0461	28.379	0.6688
	Ours	4.517	0.9847	5.489	0.0226	34.960	0.9202	8.836	0.8226	10.153	0.0452	28.540	0.6754

Table 1. Quantitative evaluation on CAVE and Pavia datasets of state-of-the-art SR methods by SAM, CC, ERGAS, RMSE, MPSNR, and MSSIM for scaling factors 4 and 8. (Best results are **highlighted**.)

projected into *query* $Q_{spa} \in \mathbb{R}^{HW \times C}$ and *key* $K_{spa} \in \mathbb{R}^{HW \times C}$. The process is defined as

$$Q_{spa} = W^Q X_{in}, K_{spa} = W^K X_{in}, \quad (8)$$

where W^Q and $W^K \in \mathbb{R}^{C \times C}$ are learnable projection matrices. Same as the operation in spectral-wise projection, we divide the projection vector into g heads. And we apply dot-product interaction on Q_{spa}^i and K_{spa}^i to generate the i -th spatial-wise projection map $A_{spa}^i \in \mathbb{R}^{HW \times HW}$,

$$A_{spa}^i = \text{Softmax} \left(\frac{Q_{spa}^i \cdot K_{spa}^i}{\sqrt{C}} + B \right). \quad (9)$$

Finally, multiplying the latent output by the spatial-wise projection matrix, we obtain the encoded feature Y_{vol} ,

$$Y_{vol} = W_{out} \cdot (Y_{spe} \cdot A_{spa} + Y_{spe}), \quad (10)$$

where $W_{out} \in \mathbb{R}^{C \times C}$ is also learnable projection matrices. Moreover, following the previous work [31], VolSA is em-

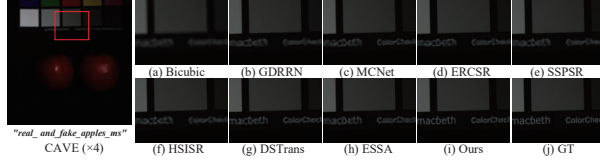


Figure 4. Visual comparison for HSI SR on the representative test image *real_and_fake_apples.ms* from CAVE dataset with spectral bands 23-15-7 as R-G-B with the scale factor 4.

played based on shift window operations by default.

4. Experiments

4.1. Super-Resolution Experimental Evaluation

Experimental Settings. In this section, we evaluate our VolFormer both qualitatively and quantitatively on natural HSI dataset, the CAVE dataset [54] and two remote sensing HSI datasets, the Pavia dataset and the Chikusei dataset [55]. DIV2K [1] is adopted for the auxiliary RGBI SR task. The results of the proposed method on the datasets are compared with seven state-of-the-art methods, *i.e.*, GDRRN[29], MCNet[26], ERCSSR[27], SSPSR[19], HSISR[23], DSTRans[56], and ESSA [62]. In the HSI SR task, we focus on upscaling factors $\times 4$ and $\times 8$ for comparison in the experiment. **All of the experiments presented in this paper are performed on NVIDIA GeForce RTX 3090 GPUs. More information about the dataset and implement details are presented in Appendix 8.1 and 8.2.**

Metrics. The quantitative evaluation of SR performance is based on six standard metrics, *i.e.*, spectral angle mapper (SAM) [58], cross correlation (CC) [32], erreur relative globale adimensionnelle de synthese (ERGAS) [43], root mean squared error (RMSE), mean peak signal-to-noise ratio (MPSNR), and mean structure similarity (MSSIM) [47].

Qualitative and Quantitative Results. The quantitative evaluation results on the CAVE, Pavia and Chikusei datasets are listed in Table 1 and Table 2, and the best results are highlighted in bold. The evaluation results demonstrate that our method outperforms the other methods in terms of upscaling factors of both $\times 4$ and $\times 8$. Especially on the CAVE dataset, our VolFormer obtains PSNR gains of 0.13 dB and 0.17 dB over the previous best method. Compared with the Chikusei and Cave datasets, Pavia has a significantly small data size. Notably, our method still maintains stable improvement of quantitative results on the Pavia dataset, which is attributed to the more comprehensive interactions, resulting in more effective feature learning. To intuitively show the performance of our method, we further exhibit the visual results. Figure 4 shows the $4\times$ super-resolution results on the CAVE dataset. Clearly, the compared methods have limitations in their effective receptive field, resulting in blurred textures. In contrast, our method has excellent

performance in terms of capturing more comprehensive interactions from both spatial and channel dimensions, which makes textures and edges clear. **More experimental results in Appendix 8.3 and 8.4.**

4.2. Denoising Experimental Evaluation

In this section, we discuss HSI denoising experiments that have been conducted on HSIs with synthetic noise and real-world noise. For the HSI denoising task on synthetic noise, different levels of Gaussian noise and mixture noise are added to the ICVL dataset [3]. For the HSI denoising task on real-world noise, we evaluate our VolFormer on a real HSI noise dataset HSDwRD [63]. RENOIR [2] is a dataset for real-world noise image denoising tasks and adopted for the auxiliary RGBI real-world denoise task. **More information about the dataset and implement details are presented in Appendix 9.1 and 9.2.**

4.2.1. Qualitative and Quantitative Results

Experimental Results on Synthetic Noise. To demonstrate the equally superior performance for HSI denoising, we compare our VolFormer with seven state-of-the-art HSI denoising algorithms, including four traditional methods, BM4D [35], KBR [49], WLRTR [6], and NGmeet [15], and four recently developed deep learning methods, including HSD-CNN [57], QRNN3D [48], DPPR [21] and SST [24]. For the denoising experiment on the synthetic noise, additive Gaussian white noise was added to each input HSI with four different noise levels, including 30, 50, 70, and random strengths ranging from 30 to 70. Besides, we add the mixture noise to the synthetic dataset. Each band is randomly corrupted by at least two kinds of Gaussian noise, impulse noise, deadlines and stripes. We evaluate the denoising performance of all methods by using two standard metrics, *i.e.*, MPSNR, and MSSIM. We compared the specific values of the quantitative indices, and the results are presented in Table 3. **The visual results can be found in Appendix 9.3.**

Experimental Results on Real-world Noise. The quantitative results can also be quantitatively corroborated by Table 4. **The visual results can be found in Appendix 9.3.**

4.3. Classification Experimental Evaluation

In this section, we implement experiments on three benchmark datasets to illustrate the effectiveness of the proposed VolSA in classification task. We use VolSA as the key component to form a classification network. Our network is built based on the Transformer-based classification network, *i.e.*, Hybrid Former [36]. We use VolSA to replace the proposed spatial-spectral attention in HybridFormer. The classifiers used in the comparison are eight state-of-the-art methods (SVM [42], DBMA [34], DBDA [28], SSRN [66], SSFTT [40], BS2T [39], CS2DT [51] and HybridFormer [36].

Scale Method	$\times 4$						$\times 8$					
	SAM	CC	ERGAS	RMSE	MPSNR	MSSIM	SAM	CC	ERGAS	RMSE	MPSNR	MSSIM
Bicubic	3.404	0.9212	6.756	0.0156	37.638	0.8954	5.016	0.8333	9.626	0.0222	34.571	0.8078
GDRRN[29]	2.807	0.9436	5.704	0.0130	39.161	0.9276	4.219	0.8675	8.676	0.0198	35.505	0.8443
MCNet[26]	3.257	0.9372	6.213	0.0138	38.323	0.9179	4.191	0.8702	8.509	0.0197	35.591	0.8509
ERCSR[27]	3.185	0.9380	6.175	0.0137	38.405	0.9194	4.182	0.8738	8.390	0.0195	35.697	0.8532
SSPSR[19]	2.409	0.9483	5.218	0.0118	39.877	0.9381	4.140	0.8770	8.373	0.0191	35.807	0.8534
HSISR[23]	2.781	0.9474	5.500	0.0126	39.594	0.9323	4.225	0.8711	8.521	0.0195	35.634	0.8463
DSTrans [56]	2.346	0.9556	4.979	0.0116	40.248	0.9427	4.073	0.8736	8.425	0.0194	35.723	0.8502
ESSA [62]	2.272	0.9591	4.785	0.0110	40.320	0.9450	4.124	0.8800	8.265	0.0188	35.819	0.8571
Ours	2.212	0.9598	4.723	0.0108	40.407	0.9461	4.024	0.8819	8.143	0.0187	36.008	0.8588

Table 2. Quantitative evaluation on Chikusei dataset of state-of-the-art SR methods by SAM, CC, ERGAS, RMSE, MPSNR and MSSIM for scale factors 4 and 8. (Best results are **highlighted**.)

Sigma	Metric	Method									
		Noise	BM4D [35]	KBR [49]	WLRTR [6]	NGmeet [15]	HSID-CNN [65]	QRNN3D [48]	DPPR [21]	SST [24]	Ours
30	MPSNR	18.589	38.451	41.478	42.622	42.988	38.704	42.217	43.056	43.322	43.551
	MSSIM	0.5523	0.9341	0.9840	0.9878	0.9889	0.9493	0.9883	0.9900	0.9923	0.9938
50	MPSNR	14.154	35.641	39.156	39.722	40.260	36.167	40.151	40.911	41.094	41.407
	MSSIM	0.3479	0.8890	0.9743	0.9781	0.9784	0.9189	0.9820	0.9843	0.9857	0.9893
70	MPSNR	11.231	33.677	36.714	37.520	38.656	34.312	38.303	38.817	39.550	39.749
	MSSIM	0.2342	0.8450	0.9605	0.9667	0.9743	0.8856	0.9742	0.9763	0.9762	0.9794
[30,70]	MPSNR	17.338	37.662	40.681	41.664	42.230	37.811	41.369	42.231	42.524	42.611
	SSIM	0.5631	0.9141	0.9790	0.9825	0.9852	0.9350	0.9847	0.9873	0.9910	0.9920
Mixture	MPSNR	10.809	20.360	26.655	12.227	17.367	31.246	27.061	35.804	36.065	36.213
	MSSIM	0.1293	0.4547	0.6378	0.1851	0.3649	0.7610	0.4465	0.9148	0.9151	0.9164

Table 3. Quantitative evaluation results of state-of-the-art denoise methods under several noise levels on ICVL dataset. (Best results are **highlighted**.)

Metric	Method									
	Noise	BM4D [35]	KBR [49]	WLRTR [6]	NGmeet [15]	HSID-CNN [57]	QRNN3D [48]	DPPR [21]	SST [24]	Ours
MPSNR	20.907	25.318	26.162	21.317	26.161	26.218	23.832	25.879	27.542	27.825
MSSIM	0.3186	0.8156	0.8279	0.5075	0.7833	0.8261	0.7917	0.5244	0.8306	0.8422
SAM	25.299	6.302	6.085	6.303	5.085	5.067	10.019	16.398	4.658	4.312

Table 4. Quantitative evaluation results of state-of-the-art denoise methods on HSIDwRD. (Best results are **highlighted**).

Method	FLOPs(G)	Params(M)	MPSNR	MSSIM	SAM
(a) SSA	78.10	4.29	39.597	0.9574	3.195
(b) CSA	78.66	4.34	39.647	0.9633	3.172
(c) SSA+CSA	93.60	5.24	40.073	0.9649	3.169
(d) VoISA w/o SLP	90.15	5.10	40.154	0.9657	3.127
(e) VoISA	90.15	5.10	40.205	0.9663	3.083

Table 5. Ablation study of different self-attention mechanisms.

Table 11 demonstrates the quantitative evaluation results for HU2013 dataset. **More results and information about the datasets can be found in Appendix 10.** The best results are highlighted in bold. The accuracy per class, OA, AA and Kappa are respectively listed. It can be seen that our method keeps the highest classification accuracy among all the methods.

4.4. Ablation Studies

Several ablation studies are conducted to analyze the contribution of the proposed VoISA and the parameter choices.

Class	SVM [42]	DBMA [34]	DBDA [28]	SSRN [66]	SSFTT [40]	BS2T [39]	CS2DT [51]	HybridFormer [36]	Ours
1	94.74±2.32	96.53±1.22	95.57±1.46	98.58±0.16	96.57±0.45	96.00±1.02	95.81±2.35	96.51±2.66	98.87±1.91
2	96.44±1.49	99.44±0.13	99.47±0.14	99.81±0.18	98.45±0.68	98.61±1.14	98.46±1.00	99.13±1.14	97.92±0.59
3	98.16±1.40	99.45±0.77	100.0±0.00	99.87±0.24	99.77±0.13	99.96±0.06	99.95±0.09	99.44±0.56	99.79±0.16
4	97.65±1.40	99.01±0.72	97.55±0.97	98.59±1.10	99.06±0.75	99.50±0.51	94.73±1.90	99.44±0.57	99.46±0.49
5	93.00±2.04	98.35±1.55	96.61±2.71	98.10±1.45	100.0±0.00	99.86±0.27	97.07±2.60	100.0±0.00	99.95±0.11
6	98.57±1.82	99.13±1.22	100.0±0.00	98.73±2.52	98.17±1.37	100.0±0.00	99.02±1.68	88.70±9.11	97.07±2.64
7	86.08±2.97	90.77±1.95	92.75±1.34	93.64±3.65	93.78±1.34	95.49±1.09	94.89±2.37	97.80±0.90	98.01±0.76
8	80.90±2.47	97.16±0.42	97.75±1.53	98.40±1.75	91.82±0.75	99.78±0.26	95.73±2.91	87.49±5.16	89.62±3.01
9	74.64±3.56	87.18±4.20	93.70±2.37	93.35±3.19	91.35±1.53	93.17±2.18	93.81±3.02	93.25±2.46	94.51±4.54
10	83.11±3.94	91.73±3.09	90.78±2.91	91.70±5.16	97.91±1.49	93.45±1.84	88.84±2.88	98.37±1.67	97.88±2.63
11	81.27±2.51	92.41±0.60	92.07±4.55	91.47±1.16	98.05±1.06	97.25±1.09	92.59±4.06	97.83±2.35	99.16±1.21
12	76.29±3.79	95.45±0.41	88.43±1.98	91.17±2.41	90.70±1.26	94.61±1.15	92.13±5.07	94.16±3.83	95.66±3.71
13	63.57±10.27	93.64±1.57	90.03±3.65	88.86±4.58	96.54±2.17	90.60±1.70	95.83±5.11	89.85±6.89	94.32±6.51
14	93.77±4.56	98.39±1.13	98.50±1.62	91.78±5.35	99.68±0.18	98.15±1.91	100.0±0.00	100.0±0.00	99.95±0.10
15	99.21±0.51	98.24±0.25	97.92±1.25	99.46±0.66	100.0±0.00	96.90±1.55	97.51±1.40	99.28±0.79	99.87±0.27
OA	87.47±0.66	95.17±0.29	94.83±0.54	95.44±0.57	96.32±0.28	96.76±0.37	94.95±0.71	96.40±0.30	97.35±0.30
AA	87.83±0.94	95.79±0.28	95.41±0.54	95.57±0.55	96.79±0.27	96.89±0.35	95.76±0.62	96.08±0.79	97.47±0.47
Kappa	86.44±0.71	94.78±0.31	94.41±0.58	95.07±0.62	96.03±0.31	96.50±0.41	94.54±0.77	96.11±0.33	97.13±0.33

Table 6. Classification results on the HU2013 dataset. (Best results are shown in bold)

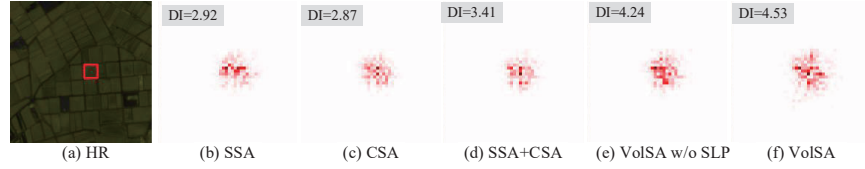


Figure 5. LAM [13] results correspond to different self-attention mechanisms. The LAM results reflect the contribution of each pixel in the degraded image when reconstructing the patch marked with a box. Diffusion index (DI) [13] reflects the interaction distance of the model. A higher DI represents a broader interaction.

All the experiments are performed on the CAVE dataset for the HSI SR task with scaling factor $\times 4$.

Discussion of the Self-Attention Mechanisms. To verify the effectiveness of the proposed VolSA, we have conducted an ablation study of several self-attention mechanisms, and the results are demonstrated in Table 5. We compare our VolSA with spatial self-attention (SSA) and channel self-attention (CSA). We also compare our VolSA with the linear combination form (SSA+CSA) of spatial self-attention and channel self-attention. Additionally, we have analyzed the effect of the proposed spectrum location prior. **VolSA w/o SLP** denotes the VolSA without spectrum location prior. Table 5 demonstrates that our VolSA provides the most competitive gain over the SSA, CSA, and the combined form. Furthermore, the spectrum location prior results in the expected improvement, and the cost of computational complexity can be negligible. As depicted in Fig. 5, the LAM [13] results and DI [13] results indicate that our VolSA has the most further interaction distance and utilizes the most pixels located over the HSI cube to restore correct and clear textures. More ablation studies and dis-

cussion about specific architecture and computational cost can be found in Appendix 11 and 12.

5. Conclusion

This paper presents an efficient Transformer, called VolFormer, for single hyperspectral image restoration, including super-resolution and denoising. VolFormer is capable of modeling more comprehensive interaction in the hyperspectral image cube without increasing largely computing costs. Specifically, we propose a particular volumetric self-attention to model 3D context information by relaying similarity matrixes in spatial and channel dimensions. Benefiting from the particular design, VolFormer transforms the complexity from cubic to quadratic. Besides, we utilize the spectrum characteristics in HSI and introduce the location prior to improve the representation ability of the self-attention mechanism. Extensive experiments for HSI super-resolution and denoising demonstrate the SOTA performance of VolFormer in terms of visual and quantitative results.

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